Rokib Al Dhin Raadh

AI Researcher · Founder & CEO, OXIEDO · Inventor of ORMAS

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AI researcher and founder, age 18. Sole author of a peer-reviewed paper on the stability of self-repairing neural networks (DeepMath 2026). Reviewer for a NeurIPS 2026 workshop and invited speaker at Cohere Labs. Developed ORMAS independently over 383 experiments on a single GPU, without institutional affiliation or external funding.

Selected Distinctions

- **Sole author, peer-reviewed paper**, DeepMath 2026 (Conference on the Mathematical Theory of Deep Neural Networks), The Ohio State University. Double-blind review; 41 papers accepted.
- **Reviewer**, NeurIPS 2026 Workshop on Trustworthy AI for Good, Paris, France.
- **Invited talk**, Cohere Labs, Toronto, Canada, November 2026.
- **Highest-ranked application**, Cosmos Institute, 2026 review cycle.
- **Accepted, top 1% of applicants**, 1752vc Ignite, 2026 (OXIEDO).

Research

Interests. Stability of neural network training under non-gradient updates; self-repairing and auditable training; input-to-state stability and switched systems.

2024–present **Independent Researcher** ORMAS: self-repairing neural networks

- Developed ORMAS, a training architecture that monitors each layer during training, repairs failing components and logs every correction with its cause.
- In the DeepMath 2026 paper, established local practical input-to-state stability under bounded repairs and showed that per-channel-bounded repairs cannot reactivate a dead BatchNorm–ReLU channel.
- After two trained convolutional layers were zeroed, ORMAS recovered to 80.3% test accuracy on CIFAR-10, compared with 10.00% for standard training (three seeds). Under sequential learning it retained 94.6% on the first task, compared with 47.3% for ResNet-18.
- Conducted 383 controlled experiments across four architecture families (DAG, two CNNs, ResNet-18) on CIFAR-10 and CIFAR-100, all reproducible from seed, on one RTX 3090. Codebase of 16,316 lines of PyTorch.
- Released ormas-monitor on PyPI, which implements the dead-channel result as a runtime check. Applied to 22 torchvision models, it found near-zero channels in 19 of the 21 that use BatchNorm and ReLU; removing them from VGG-19-BN reduced compute by 18.6% and changed none of its top-1 predictions on 10,000 ImageNetV2 photos.

Publications

Peer-reviewed

[1] **R. A. D. Raadh**. "Self-Repair as a Bounded Disturbance: Input-to-State Stability of Neural Network Training Dynamics." *DeepMath 2026: Conference on the Mathematical Theory of Deep Neural Networks*, The Ohio State University, 2026. Double-blind review; poster. OpenReview · Poster

Preprints and technical reports

[2] **R. A. D. Raadh**. "ORMAS: Neural Architectural Transparency Enables Autonomous Self-Correction." Preprint, Zenodo, 2026, doi:10.5281/zenodo.21730363. With a 36-page supplement and code that reproduces all 383 experiments.

[3] **R. A. D. Raadh**. "OXIMO/AX09: Autonomous LLM-Dependent Commerce. A Twelve-Month Controlled Ablation Study." Technical report, 2026.

Research proposals

[4] **R. A. D. Raadh**. "Can a Machine's Own Failure Record Be Read? Evaluating the Human Usability and Reliability of Automatically Generated Neural Network Incident Logs." Research proposal, International Academic Research Competition (IARCO 2026), 2026. Finalist.

Invited Talks and Presentations

Nov 2026 **Invited talk**, Cohere Labs, Toronto, Canada.
When the Update Is Not a Gradient: Input-to-State Stability of Self-Repairing Neural Network Training.
Recording · Slides

Oct 2026 **Poster**, DeepMath 2026, The Ohio State University, Columbus, OH.
Self-Repair as a Bounded Disturbance: Input-to-State Stability of Neural Network Training Dynamics.

Professional Service

2026 **Reviewer**, NeurIPS 2026 Workshop on Trustworthy AI for Good, Palais des Congrès de Paris, France. Submissions on mechanistic interpretability, attribution, model auditing and post-deployment monitoring. Workshop · Reviewer list

Entrepreneurship

2023–present **Founder & CEO, OXIEDO** Auditable AI training for regulated industries

- Licenses ORMAS for on-premise use in regulated industries. Each training run produces a tamper-evident record of model changes, for institutions such as banks and hospitals that cannot yet train on their own data.
- Accepted to 1752vc Ignite (top 1% of applicants) and offered a place in the Freshmango accelerator. Finalist, The Bridge by Entrepreneur First, San Francisco. Invited to the Onstage W26 pre-pitch. Investment discussions ongoing with 500 Global and 1752vc.

2023–2026 **Sole Architect, OXIMO** Multi-agent operating system

- Designed and built a multi-agent system of 40,933 lines of Python with 2,069 tests, refactored from a 106,000-line codebase. Agents create new specialist roles as tasks require.

2025–2026 **Founder & CEO, Black Bloxie LTD** England & Wales, company no. 16711223

- Ran a twelve-month controlled study of OXIMO on a live e-commerce business. Output fell 91% when the system was removed and exceeded its original level after it was restored.

2020–2025 **Five earlier ventures**, ages 12 to 17: a tournament gaming app (co-founder), the EWM trading algorithm, a web-development sales agency, an e-commerce store, and a blockchain software project (discontinued). Sold the EWM algorithm to a U.S. client for about $10,000 at age 15.

Honors and Awards

2026 **Finalist**, International Academic Research Competition (IARCO 2026), for research proposal [4]. Ranked 24th of 145 in the Junior division, from about 500 participants in 60 countries. Results

2026 **Highest-ranked application**, Cosmos Institute, 2026 review cycle. Invited to the technical track.

2026 **Research community member**, Cohere Labs, Toronto, Canada. Invited on the basis of ORMAS, which the lab described as showing "remarkable initiative in ML safety and auditability."

2026 **Member**, FIAP, São Paulo, Brazil. Sponsored for MIT Open Learning's Universal AI program.

Software

2026 **ormas-monitor** (PyPI). Training monitor for PyTorch with per-layer alarms, dead-channel certification, optional repair and a tamper-evident log. Documentation

2026 **ORMAS reference implementation**. Reproduces all 383 experiments. Source · Run archive

Education

present **High school senior; prospective undergraduate.** Homeschooled, with senior-year high school coursework. Independent study in machine learning, control theory and the mathematics of learning dynamics.

2025–2026 **MIT Open Learning (MITx Online)**. Seven programs completed in foundation models, AI in medicine, energy, transportation and entrepreneurship.

2025–2026 **DeepLearning.AI**. Deep Learning Specialization (five courses).

Additional

Nov 2024 Guinness World Record attempt. 28 unedited livestreamed work sessions over two weeks.

Languages: English (fluent), Bengali (native).